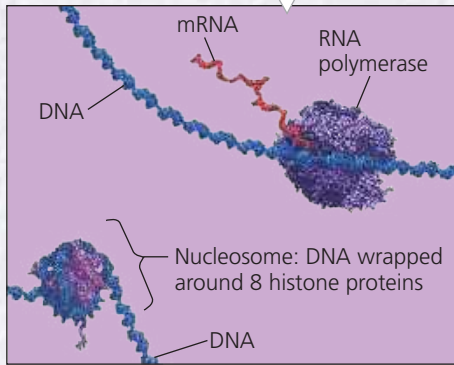


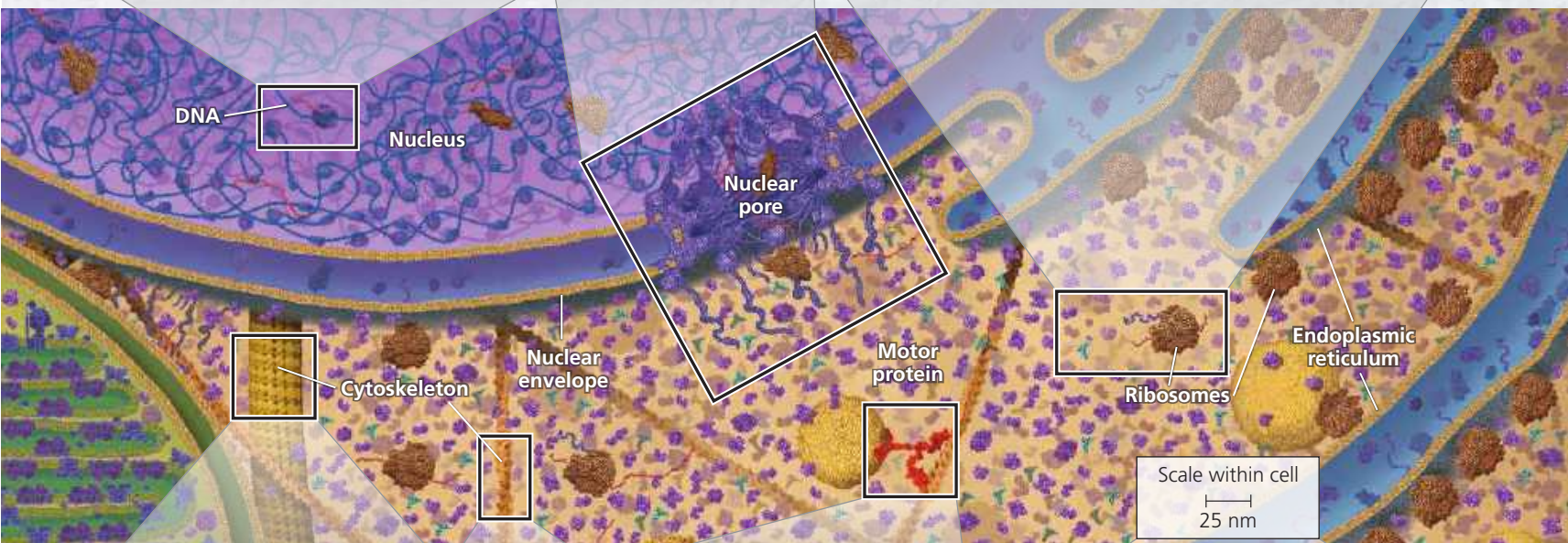
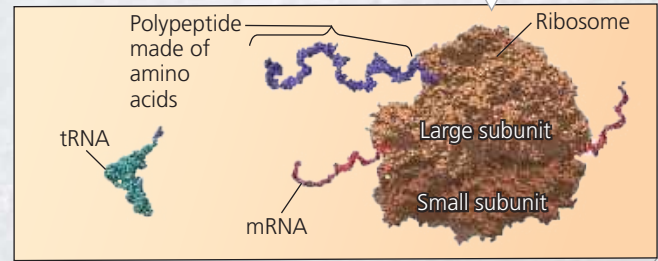
(d) Transcription (Chapter 17) In the nucleus, the information contained in a DNA sequence is transferred to messenger RNA (mRNA) by an enzyme called RNA polymerase. After their synthesis, mRNA molecules leave the nucleus via nuclear pores.



(e) Nuclear pore (Concept 6.3)

The nuclear pore complex regulates molecular traffic in and out of the nucleus, which is bounded by a double membrane. Among the largest structures that pass through the pore are the ribosomal subunits, which are built in the nucleus.

(f) Translation (Chapter 17) In the cytoplasm, the information in mRNA is used to assemble a polypeptide with a specific sequence of amino acids. Both transfer RNA (tRNA) molecules and ribosomes play a role. The eukaryotic ribosome, which includes a large subunit and a small subunit, is a colossal complex composed of four large ribosomal RNA (rRNA) molecules and more than 80 proteins. Through transcription and translation, the nucleotide sequence of DNA in a gene determines the amino acid sequence of a polypeptide, via the intermediary mRNA.

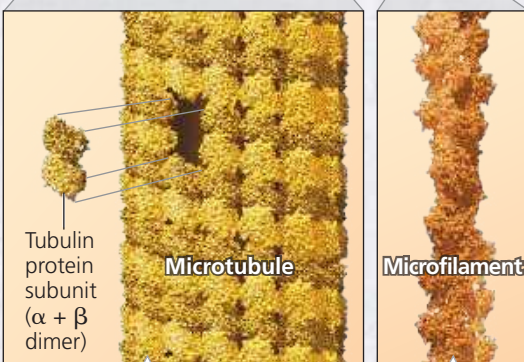


Scale within cell
25 nm

25 nm
Scale of enlarged structures

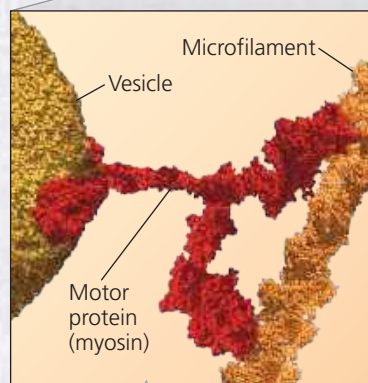
(g) Cytoskeleton (Concept 6.6)

Cytoskeletal structures are polymers of protein subunits. Microtubules are hollow structural rods made of tubulin protein subunits, while microfilaments are cables that have two chains of actin proteins wound around each other.



(h) Motor proteins (Concept 6.6)

Motor proteins, such as myosin, are responsible for transport of vesicles and movement of organelles within the cell.



1. List the following structures from largest to smallest: proton pump, nuclear pore, cyt c, ribosome.
2. Considering the structures of a nucleosome and of RNA polymerase, speculate about what must happen before RNA polymerase can transcribe the DNA that is wrapped around the histone proteins of a nucleosome.
3. Find another myosin motor protein walking on a microfilament in this figure. What organelle is being moved by that myosin protein?

➔ **Instructors:** Additional questions related to this Visualizing Figure can be assigned in Mastering Biology.